Motors and generators

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

SUCCESS CRITERIA - I can...

1. Trace energy through a motor and a generator.
2. Identify magnetic interactions in rotation or induction.
3. Include less useful energy transfers.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A consistently drawn magnetic-field map has 3 times as many field lines through the same small perpendicular area at A as at B. What does the denser spacing indicate?

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\eta = P_{useful}/P_{input}$; $E_{input} = E_{useful} + E_{losses}$. Motor: electrical to mechanical; generator: mechanical to electrical. Count losses and any stored-energy change.

Worked reasoning: An idealized motor receives 20 J electrically and delivers 12 J mechanically; 8 J goes to other transfers. A generator runs the energy conversion in the opposite useful direction.

Your turn: Draw separate motor and generator energy-flow diagrams.

Marzano check - circle: ① ② ③

03 Look and explain

Complete two energy chains: battery → motor → turning axle, and turning axle → generator → lamp. Label each input and useful output.

Marzano check - circle: ① ② ③

04 Build the solution

A motor receives 100 J electrical energy and outputs 60 J useful mechanical energy. Find efficiency and energy in other forms.

Marzano check - circle: ① ② ③

05 Use the idea

A motor drives a generator that feeds the motor. Explain why real losses prevent the loop from running forever without external input.

Marzano check - circle: 1 2 3

06 Choose a method

An electric field of 6 N/C points left. Draw the force direction on +4 C and on -4 C, then compare sizes.

Marzano check - circle: 1 2 3

07 Exit check

Which device converts mechanical input into electrical output?

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A motor draws the same electrical power in two designs, but one delivers more mechanical power. Compare efficiencies and name a measurement needed.

Marzano check - circle: 1 2 3